C10 2D Lighting

CRE132 worksheet

What you'll make

A forest clearing at night. A dim blue light covers everything, flickering torches glow orange, a moonbeam shines down into the clearing, and the player carries a lantern. Trees and rocks cast shadows that swing round as the player walks past.

You'll use Unity's **2D lights**: the **Global** light that sets the base brightness, **Spot Light 2D** for round lights and cones, **shadows** with **Shadow Caster 2D**, and **Target Sorting Layers** to choose what each light touches. No new code: the player uses C04's and C07's scripts.



Figure 1: The finished scene: a dark blue forest with a lit clearing, three torches and the player in the middle.

Before you start

1. Unzip `CRE132_C10_2DLighting_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_C10_2DLighting_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Core/C10_2DLighting`.
3. Press **Play** and walk around with WASD or the arrows. Everything is evenly lit, like daytime, and nothing casts a shadow. Press Play again to stop.

The scene has the forest clearing (**Grid**), eight trees and three rocks inside **Scenery**, three animated **Torches**, the **Player** (C04's and C07's scripts) and a **Global Light 2D**. There are no other lights yet: you'll add them all.

1. The 2D Renderer and the Global light

This project uses URP's **2D Renderer** (`Assets/Settings/Renderer2D`), which is what makes 2D lights work. Sprites use the **Sprite-Lit-Default** material, so lights affect them: click a tree and look at its Sprite Renderer's **Material**.

Select **Global Light 2D** in the Hierarchy. Like every earlier sample, it's white with **Intensity** 1, which lights everything evenly. In its **Light 2D** component, set:

- **Color:** a dark blue (R 0.45, G 0.55, B 1)
- **Intensity: 0.25**

Test it: the scene goes dim and blue, like night. Now untick the Global Light 2D's checkbox (next to its name): everything goes black, because a lit sprite with no light on it is black. Tick it again.

2. Torch lights

Add it: right-click one **Torch** in the Hierarchy and choose **Light > Spot Light 2D**. It becomes a child of the torch, so it moves with it. Rename it **Torch Light** and set its **Position** to 0, 1.6, 0, which puts it on the flame. Set:

Setting	Value
Color	orange (R 1, G 0.6, B 0.25)
Intensity	1.2
Radius: Inner / Outer	0.9 / 4.5
Falloff Strength	0.5

- **Inner Radius** is fully bright; the light fades out between Inner and Outer. **Falloff Strength** changes how quickly it fades.
- A Spot Light 2D with its angles at 360° shines all the way round, like a bulb.

Copy the light to the other two torches: select **Torch Light**, press **Ctrl+D** to duplicate it, and drag the copy onto the second **Torch** in the Hierarchy. Set its **Position** back to 0, 1.6, 0. Do the same for the third **Torch**.

Test it: press Play. Each torch makes a warm pool of light, and the flickering flame sprite still animates.

3. A moonbeam: narrowing the angle

Add it: choose **GameObject > Light > Spot Light 2D** from the menu bar. Rename the new light **Moonbeam (Spot Light)**. Set:

Setting	Value
Color	pale blue (R 0.7, G 0.8, B 1)
Intensity	0.7
Radius: Inner / Outer	1.8 / 9
Inner / Outer Spot Angle	30 / 60
Position	0, 6.5, 0
Rotation Z	180

The light shines along the object's **up** axis (the green arrow). Rotating it 180° points it straight down into the scene from above.

Test it: a cone of pale light shines down into the clearing from the top. Try other **Rotation Z** values to aim it, then set it back to 180.

4. A lantern that follows the player

Add it: right-click the **Player** in the Hierarchy, choose **Light > Spot Light 2D**, and rename it **Lantern**. Check its **Position** is 0, 0, 0. Because it's a child of the player, it moves with them. Set:

Setting	Value
Color	warm white (R 1, G 0.92, B 0.75)
Intensity	1
Radius: Inner / Outer	0.7 / 3.5

Test it: press Play and walk around. The light comes with you.

5. Shadows

Shadows need two things: a **light with Shadows turned on**, and **objects with a Shadow Caster 2D**.

On the lights: on each of the three **Torch Lights**, tick **Shadows** in the **Light 2D** component and set its **Strength** to **0.8**. On the **Lantern** and the **Moonbeam**, tick **Shadows** with **Strength 0.75**.

On the objects: expand **Scenery** in the Hierarchy, click the first object inside it, then Shift-click the last so all eleven trees and rocks are selected. Click **Add Component** in the Inspector and choose **Shadow Caster 2D**. With all eleven still selected, set:

- **Casting Source: Sprite Renderer**, so the shadow uses the sprite's shape.
- **Casting Option: Cast Shadow**, so the tree throws a shadow but its own sprite stays lit.

Test it: press Play and walk round a tree. Its shadow swings away from the lantern, and the torches throw shadows too.

6. Trim Edge

Look closely at the small rocks: they may throw a dark square, because their sprite's shape is nearly the whole square around them. Select all eleven trees and rocks in **Scenery** again, and on the **Shadow Caster 2D** raise **Trim Edge** to **0.2**. The shadow shape shrinks to fit each sprite. (The Finished scene uses 0.2 on every tree and rock.)

Test it: press Play and walk past the rocks. Their shadows follow the rock's shape, with no dark squares.

7. Target Sorting Layers

Each light chooses which sorting layers it lights. Select one **Torch Light**, open the **Target Sorting Layers** drop-down in its **Light 2D** component and untick **Characters**.

Test it: walk the player into that torch's light. The ground and trees are lit, but the player stays dark. Tick **Characters** again.

Check your work

- ☐ The scene is dim blue, and switching the Global light off makes it black.
- ☐ All three torches glow orange, and each light is a child of its torch.
- ☐ The moonbeam is a cone pointing down.
- ☐ The lantern follows the player.
- ☐ Trees and rocks cast shadows away from the lights, with no dark squares.
- ☐ Compare with `CRE132_C10_2DLighting_Finished` if something looks different.

Key points

- **No lights means black sprites** (with lit materials). The Global light sets the base brightness.
- **Lights are components**, so they can be parented (the lantern), animated and turned on and off from code.
- **Shadows need both** a light with Shadows on and objects with a Shadow Caster 2D.
- **Normal maps** (optional): with a normal map on a sprite (Sprite Editor, **Secondary Textures**), lights give it a 3D look. The art packs used here don't include normal maps.

If something goes wrong

- **Everything is black:** the Global Light 2D is off or deleted, or its Intensity is 0.
- **A light does nothing:** check its **Target Sorting Layers** include the layers your sprites are on, and that its Intensity and Outer Radius aren't 0.
- **No shadows:** the light's **Shadows** box isn't ticked, or the object has no **Shadow Caster 2D**.
- **The tree itself goes dark:** its Casting Option includes **Self Shadow**. Use **Cast Shadow**.
- **The moonbeam shines upwards:** set its Rotation Z to 180.
- **A torch's light is in the wrong place:** it isn't a child of that **Torch**, or its **Position** isn't 0, 1.6, 0.
- **The lantern stays behind when the player walks:** it isn't a child of the **Player**. Drag it onto the Player in the Hierarchy.

Try it

1. Make a torch flicker: a script that changes the light's `intensity` a little each frame with `Random.Range`. The `Light2D` class needs `using UnityEngine.Rendering.Universal;`.
2. Drag the Global light's Intensity slowly up to 1 and back down in the Inspector while playing: a day and night fade.
3. Give the player a Shadow Caster 2D too. What does the lantern do then?